

A Randomized Trial of a Computer-Tailored Decision Aid to Improve Prostate Cancer Screening Decisions: Results from the *Take the Wheel* Trial

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Abstract

Objective: To evaluate a decision aid (DA) designed to promote informed decision making for prostate cancer screening.

Methods: Twelve work sites were randomly assigned to an intervention or nonintervention comparison condition. Intervention sites received access to a computer-tailored DA at the workplace. Male employees age 45 years and above ($n = 625$) completed surveys at baseline and at 3-month follow-up, documenting aspects of informed decision making.

Results: Using an intention-to-treat analysis, men in the intervention group were significantly more likely to have made a screening decision and to have improved knowledge without increased decisional conflict, relative to men in the comparison group. These changes were observed despite the fact that only 30% of men in intervention sites used the DA. Among DA users, similar improvements were observed, although the magnitudes of changes were substantially greater, and significant improvements in decision self-efficacy were observed.

Conclusions: A DA offered in the workplace promoted decision making, improved knowledge, and increased decision self-efficacy among users, without increasing decisional conflict. However, participation was suboptimal, suggesting that better methods for engaging men in workplace interventions are needed.

Impact Statement: This trial shows the efficacy of a computer-tailored DA in promoting informed decisions about prostate cancer screening. The DA was delivered through work sites, thereby providing access to resources required to participate in informed decision making without requiring a medical appointment. However, participation rates were suboptimal, and additional strategies for engaging men are needed. *Cancer Epidemiol Biomarkers Prev*; 19(9); 2172–86. ©2010 AACR.

Introduction

Prostate cancer (CaP) is the most commonly diagnosed cancer among men in the United States. Established risk factors (age, family history, black race) are not modifiable (1). Therefore, cancer control efforts have focused on early detection with the prostate-specific antigen (PSA) test for men ages 50 years and over. However, PSA screening remains controversial, and data from two long-awaited trials have only intensified debate about the role of routine screening (2, 3).

Until recently, routine screening for CaP was recommended only by a few organizations, including the American Cancer Society (4) and the American Urological Association (5). Other medical organizations advised men to learn about the potential benefits, limitations, and harms of the test, through a process termed “informed decision making” (IDM; refs. 6–9). In 2010, the American Cancer Society revised their recommendations, emphasizing the importance of informed decision making rather than mass screening (4). According to the U.S. Preventive Services Task Force (USPSTF), an informed decision is one in which an individual: (a) understands the nature of the disease being addressed as well as the risks, limitations, and benefits of clinical services (that is, adequate knowledge); (b) is confident in his ability to participate in decision making at a personally desired level (decision self-efficacy); and (c) has considered his personal preferences and makes a decision consistent with his values (decisional consistency; ref. 10).

Recently, there has been a proliferation of interventions to promote IDM, particularly in the form of decision aids (DA). DAs are tools designed to help individuals make

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choices by providing pertinent information, presenting data about the likelihood of potential outcomes, and elucidating personal values associated with each option, with a goal of promoting decisional consistency (11, 12). There are few published trials of computer-based DAs for CaP screening (13-16) and, with one exception (15), have all been conducted in clinical settings. The present study was conducted in work sites, a setting that affords access to a large segment of men who are age appropriate for IDM about CaP screening, has existing communication channels that can facilitate promotion efforts, and offers an infrastructure through which intervention activities may be institutionalized and sustained. By targeting particular industries, it is also possible to gain access to workers in particular occupations. We recruited manufacturing industries with a goal of including high proportions of workers, who, by nature of their income and educational level, may have diminished access to information about CaP screening.

The purpose of this randomized trial was to evaluate the efficacy of a computer-tailored DA designed to promote IDM for CaP screening among employed men. Primary outcomes were decisional status (decided/undecided) and individual components of IDM as defined by the U.S. Preventive Services Task Force: knowledge, decision self-efficacy, and decisional consistency (6). Secondary outcomes included desire for involvement in decision making (control preferences) and decisional conflict. We hypothesized that at the conclusion of the study, relative to men in comparison work sites, men in intervention work sites would (a) be more likely to have made a screening decision and (b) show significantly higher levels of knowledge, decision self-efficacy, and decisional consistency. Further, we anticipated that men in the intervention group would desire greater involvement in the decision-making process and would report lower decisional conflict.

Materials and Methods

In this trial, work sites were the unit of randomization and intervention, whereas individual employees constitute the unit of measurement. Twelve work sites were randomly assigned to a 3-month intervention or to a non-intervention comparison group. Using an intention-to-treat analysis, the efficacy of the DA was evaluated by comparing mean changes in decisional status (decided/undecided) and IDM variables between baseline and follow-up in intervention and comparison sites. A process-tracking system documented intervention delivery and characteristics of DA users.

Setting

The Dun & Bradstreet database was used to identify work sites with Standard Industrial Classification codes 20 to 39 that represent manufacturing industries. Additional criteria for site selection included the following: (a) ≥ 100 men in the targeted age range (45-70 y), (b)

$<20\%$ employee turnover in the year before study initiation, and (c) location within 90 minutes of the study center (Boston, MA). A total of 161 companies were contacted to determine eligibility, which was assessed through a five-page mailed survey documenting organizational and workforce characteristics. Seventy-one companies (44%) did not return the survey; 74 companies (46%) completed the survey but were ineligible ($n = 45$ due to size). Four sites ($n = 3\%$) declined participation, citing layoffs, acquisitions, or restructuring as reasons for nonparticipation (see Fig. 1). The 12 participating work sites ranged in size from 100 to 1,000 employees (mean = 650). Sites were blocked on size (total employees <500 ; ≥ 500) and percent of male employees ($<50\%$; $\geq 50\%$), and randomly assigned by computer-generated random numbers to condition within blocks.

Intervention development

The DA development process followed a strategy described by Bartholomew et al. (17) for the development of theory-based interventions. The first step was articulation of a conceptual framework. We employed the Ottawa Decision Support Framework, which identifies factors that influence decision making and are amenable to modification through decision support (12). The Decision Support Framework also suggests a step-by-step approach for decision making: identification of available options, acquisition of information necessary for decision making, clarification of values relevant to the decision, and development of a plan for action.

Content of the DA was based on expert opinion about information necessary for IDM (10, 18), as well as guidelines from the International Patient Decision Aid Standards. The International Patient Decision Aid Standards is an international group of experts in the field of decision making that has developed guidelines for the creation of high-quality DA tools (19). Criteria include the provision of (a) balanced information about various options based on evidence, (b) information about the probabilities of potential outcomes associated with each option, (c) exercises to elucidate values relevant to available options, and (d) guidance about the development of a plan that will facilitate progress toward the chosen option. Factual content considered "required" for IDM included the following: prevalence of CaP; CaP risk factors; methods used for early detection and their operating characteristics (that is, sensitivity, specificity); potential advantages and disadvantages of screening, and men's evaluation of the relative importance of these factors; the recommendation of major medical organizations that men make individualized decisions; and meaning of an elevated PSA test and methods for diagnosis. After viewing the aforementioned required IDM content, men were able to select from a menu of additional topics (e.g., treatment options) according to their interests.

During the development process, focus groups ($k = 4$; $n = 29$) were conducted to gauge men's responses to

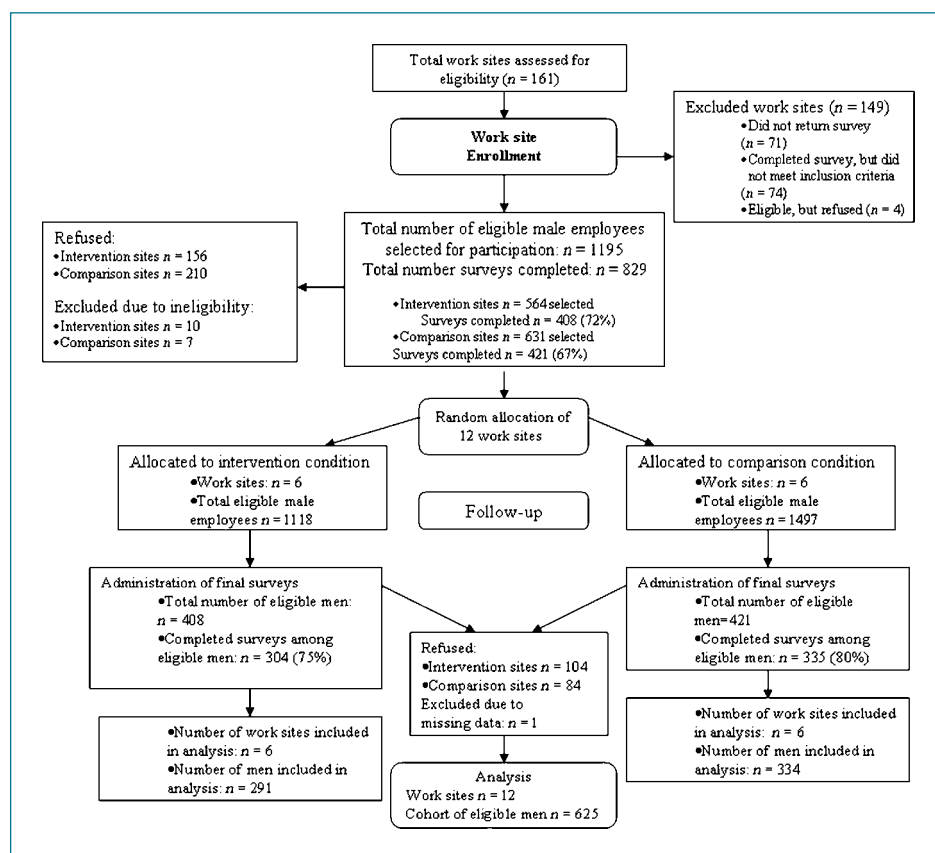


Figure 1. Sampling schema—work sites and employees.

educational messages and to assess their reactions to various communication strategies. All participants were recruited from work sites not participating in the trial. During these discussions, several themes became evident, including the importance of taking charge of one's health, making independent decisions, and masculinity. In addition, some men used a "road map" analogy to communicate about the various decision points in screening. Therefore, we conceived of the study name "Take the Wheel," to reflect a masculine theme in which men were taking an active role in decisions about their own health care and "steering their own course" (that is, making decisions). Based on focus group feedback, a functional version of the computer DA was developed and tested among men ($n = 10$) recruited from nonparticipating work sites to assess acceptability of the graphics, actors, narrator, and script. Following revisions based on audience feedback, the prototype was further refined. Usability testing was conducted with an additional 15 men recruited from nonparticipating sites to assess navigability, efficiency of use, error frequency and severity, and satisfaction among end users. The final DA consisted of interactive video and audio components, with a computer touch screen and minimal on-screen text.

Based on our conceptual model and prior work, the DA was tailored on three characteristics: personal risk

for CaP, individual ratings of the pros and cons of screening, and decisional consistency. Personalized risk was calculated using the YourDiseaseRisk algorithm (20). Men input data in response to questions about risk factors and were provided with an on-screen graphic of their risk relative to other men their age (that is, greater than average, average, less than average). In addition, men were asked to weigh the pros and cons of screening. This information was then pictorially presented as a balance scale, with pros on one side and cons on the other. Men were then asked about their screening preference ("decided to be screened," "undecided," or "decided not to be screened"). If their rating of the pros/cons was not compatible with their screening decision, the user was informed that these positions were "inconsistent." In this case, the individual was encouraged to return to the menu of topics to learn more, to take more time to think things over, or to discuss their decision with their health care provider or significant other. To promote self-efficacy with respect to decision making, the DA presented several scenarios with different men going through the decision-making process to role model a variety of decision-making approaches. Men were subsequently coached through the steps of decision making, as outlined by the Decision Support Framework. Additional information about the DA is available elsewhere (21).

Intervention delivery

Between November 2006 and June 2007, the DA was delivered in intervention sites through tablet computers made available in common gathering areas (e.g., break rooms, cafeterias). We elected to make computers available in public spaces, with the assumption that visibility would generate interest and, thus, promote participation. Each location afforded sufficient privacy so that DA users could sit individually and view the computer screen without their responses being seen by others. Headphones were provided. The DA was designed to be independently administered even for those with minimal or no computer skills. A health educator was available to provide assistance with computers if needed, although no individual required assistance, other than initial start-up of the program.

We used multiple strategies to publicize and promote the intervention, including posters placed in high-visibility areas, distribution of fliers, announcements made at regularly scheduled meetings, and provision of small incentives (e.g., key ring flashlights). Computers were made available in work sites at prespecified days, based on agreements between management and study staff. The computers were available during the day, generally in 6-hour periods, based on managements' request. Each site had at least three computers available on site for a minimum of 15 days over the 3-month intervention period (roughly once per week). Men were allowed release-time from work to use the DA. Information was saved at each time of use; men could either complete the DA session at one time or return at multiple time points to complete it (mean time spent, 28 min). At the conclusion of the session, men were provided with a printed tailored report summarizing their estimated risk for CaP, assessment of pros/cons, decisional status, and pages visited during DA use. This report was designed to facilitate communication about screening with primary care providers.

Data collection

Baseline data were collected through self-administered pencil-and-paper surveys between September 2006 and March 2007; follow-up assessments were conducted between March 2007 and July 2007. Those eligible were men age ≥ 45 years who were permanent employees working ≥ 20 hours per week. Forty-five years was selected as a minimum because some medical organizations advise that men be offered screening at < 50 years if they are at higher-than-average risk for CaP (22). Men employed temporarily, or those who worked less than half-time, were excluded from data collection because these factors would diminish access to the intervention. Employee rosters were used to identify eligible men. In sites that had > 100 eligible men, a random sample was selected. In sites with 100 men, a census was surveyed.

Eligible men were sent a letter inviting them to participate and that provided informed consent information. Two weeks later, surveys were distributed and collected

through company mail or by hand by study staff. The survey cover sheet reiterated informed consent information; completion of surveys was taken as consent to participate. All written communication was at the sixth-grade level. Men were allowed to complete the survey on site during work hours and were provided a financial incentive (\$25) after each time point. Nonrespondents were contacted up to three times by electronic mail, telephone, or in person. Across the 12 sites, the mean baseline response rate was 72% ($n = 812$; range = 59-86%). At follow-up, the mean response rate among those who completed the baseline survey was 79% ($n = 639$; range = 61-85%), resulting in a final cohort of 625 men. All procedures were approved by Institutional Review Board at the Dana-Farber Cancer Institute.

Measures

Primary outcomes

Decisional status. The validated Stage of Decision Making Scale (23) asked respondents to rate their readiness to make a decision, with five response options ranging from "I haven't thought about it before" to "I have made a decision and I am not likely to change my mind." Men were classified as having "decided" if they stated either that they had made a decision, but were willing to reconsider, or if they responded that they had made a decision, but were unlikely to change their mind. Those "undecided" reported that they had not thought about the decision, or were uncertain.

CaP knowledge. Men's recognition of the PSA test was assessed using a standard single item: "The prostate specific antigen test (PSA) is a blood test that is used to find prostate cancer. Before now, had you ever heard of the PSA test?" (yes/no). Fourteen validated questions assessed men's knowledge of CaP prevalence, risk factors, screening modalities, diagnostic procedures, and treatment-related complications (24). The proportion of accurate responses was transformed to a percentage scale ranging from 0% (no correct responses) to 100% (all correct responses). The internal reliability in our sample was adequate (Cronbach's $\alpha = 0.69$).

Decision self-efficacy. The validated 11-item Decision Self-Efficacy Scale with three response categories was used to assess confidence in one's ability to participate in decision making to the extent desired (25). Respondents were asked to reflect on their confidence level about various aspects of the decision-making process, with response options of "very confident" (score = 4) to "not at all confident" (score = 0). Scores were summed, divided by 11, and multiplied by 25, to arrive at a range of scores from 0 (no self-efficacy) to 100 (higher self-efficacy). In this sample, the internal consistency coefficient was high ($\alpha = 0.91$).

Consistency between values and screening decision. We assessed the congruence between screening preference and personal values relevant to the screening decision, an approach similar to that used by Sepucha et al. (26) in a recent study of breast cancer treatment decisions. First, to

assess screening preference, men were asked "If you had to decide now, what would you choose?" Options included "to get a PSA test," "not to get a PSA test," and "I could not decide." In the literature, assessment of values has primarily been measured with probability-based risk-benefit trade-offs (27). We pretested these items in focus groups ($k = 1$; $n = 15$) and found them unacceptable to a majority of men. Therefore, we developed items to assess the personal importance or relative worth of the advantages and limitations of screening, based on focus groups themes and published literature (e.g., importance of information, accuracy of test, potential side effects of treatment). Further information about scale development is available elsewhere (28). Individual items are presented in Appendix 1. High positive scores reflect values strongly in favor of screening, and high negative scores reflect low importance of screening (range, +16 to -16). Principle components analysis identified a single factor, providing support for combining all items in one scale. The internal reliability of the values questions was good ($\alpha = 0.81$). In test-retest assessments 6 months apart, men ($n = 812$) rated their values consistently (concordance coefficient = 0.682; ref. 28).

Based on a receiver operating characteristic curve of the sensitivity and specificity of responses to questions related to values, we set the cutoff for favorable versus unfavorable value ratings of screening at "zero." An individual's screening decision was considered "consistent" with their values if they reported an intention to be screened *and* their values score was greater than zero. Having a negative value score and lack of intention to be screened was considered consistent. Those considered "inconsistent" had value ratings that did not align with screening preference.

Secondary outcomes. *Preference for Control in Decision Making* was assessed through the Control Preference Scale (29) with a single item. Individuals were asked "Who should make medical decisions?" Response options included the following: (a) "I make the final decision about screening on my own"; (b) "I make the decision after seriously considering my doctor's opinion"; (c) "My doctor and I share responsibility for the decision"; (d) "I prefer that the doctor make the decision after seriously considering my opinion"; and (e) "I prefer that the doctor make the decision." In analyses, responses were collapsed to reflect active decision-making styles (options a and b), collaborative styles (option c), and passive styles (options d and e; ref. 29).

Decisional Conflict was measured through the validated Decisional Conflict Scale (12). Respondents were asked to rate statements such as: "The decision I made was the best decision possible for me personally." Responses were gauged on a five-point scale ranging from "strongly agree" to "strongly disagree." Scales were standardized from 0 (no conflict) to 100 (extreme conflict). In this sample, internal consistency was good ($\alpha = 0.84$).

Sociodemographic characteristics, health behaviors, screening history, and access to health care were also as-

sessed, using standard items from the CDC's Behavioral Risk Factor Surveillance Surveys.

Statistical analysis

The work site was the unit of recruitment and intervention. All analyses incorporated the clustering of respondents within work sites (intraclass correlation coefficient = 0.0068) using generalized estimating equations to fit generalized linear models. An intention-to-treat approach was used in all analyses, using all available data. P values were two sided, and a P value of <0.05 was termed significant. With this sample size, assuming a type one error of 0.05, the design had 79% power when the coefficient of variation (ratio of SD to mean) was 1.2 (30). All analyses were carried out using SAS statistical software, version 9.1 (SAS Institute).

Discrete variables, such as age group, race/ethnicity, and marital status, were compared using χ^2 tests. The Wilks-Shapiro test was used to assess the normality of continuous variables. Linear regression was used to examine mean change between baseline and follow-up for the outcomes of knowledge, decision self-efficacy, and decisional conflict across intervention and comparison groups. Logistic regression was used to examine follow-up values of the outcomes of decisional status, decisional consistency, and control preferences, controlling for baseline values across intervention and comparison groups.

Control variables included age, race/ethnicity, income, education, marital status, family history of CaP, and previous PSA test. We included some of these variables although they were not significantly associated with outcomes at the $P > 0.05$ level in bivariate analyses, with a goal of controlling for potential confounding (that is, change in coefficients of $\geq 10\%$). However, some outcomes had limited variability (that is, decisional consistency and control preferences). In these cases, we were unable to fit adjusted models with all of the aforementioned control variables. Adjusted analyses for decisional status include only age and education. Adjusted models were not calculable for control preferences or decisional consistency, due to limited variability in responses.

Results

Characteristics of cohort

Table 1 summarizes characteristics of the samples at baseline and follow-up, by intervention group. Overall, the cohort was characterized by a slightly larger percentage of non-Hispanic white men, compared with the total baseline sample (92% versus 89%, $P < 0.01$). Other sociodemographic and health characteristics were not significantly different between baseline and follow-up.

Changes in primary outcomes by treatment group

Table 2 presents bivariate relationships among sociodemographic and health characteristics across the range of primary and secondary outcomes. Older men were more likely to show decisional conflict. White race was associated with decreased decisional self-efficacy. Those

Table 1. Demographic characteristics of samples at baseline ($n = 812$) and follow-up ($n = 625$) by intervention arm, Take the Wheel Trial

Characteristic	Baseline ($n = 812$)		P^*	Follow-up ($n = 625$)		$P^\dagger$
	Intervention	Comparison		Intervention	Comparison	
	$n = 398$ n (%)	$n = 414$ n (%)		$n = 291$ n (%)	$n = 334$ n (%)	
Age			0.98			<0.01
45-49	151 (38)	120 (29)		116 (40)	96 (29)	
50-54	106 (27)	126 (30)		78 (27)	104 (31)	
≥ 55	93 (23)	153 (37)		64 (22)	126 (38)	
Missing	48 (12)	15 (4)		33 (11)	8 (2)	
Race/ethnicity			<0.01			0.41
White, non-Hispanic	354 (89)	372 (90)		259 (89)	309 (93)	
Other	37 (9)	39 (9)		26 (9)	22 (7)	
Missing	7 (2)	3 (1)		6 (2)	3 (1)	
Household Income			0.37			0.01
<\$50,000	65 (16)	59 (14)		43 (15)	46 (14)	
\$50,000-74,999	117 (29)	80 (19)		86 (30)	63 (19)	
$\geq$ \$75,000	195 (49)	247 (60)		146 (50)	201 (60)	
Missing	21 (5)	28 (7)		16 (5)	24 (7)	
Education			0.08			0.05
High school or less	131 (33)	74 (18)		93 (32)	55 (16)	
Some college	129 (32)	141 (34)		94 (32)	123 (37)	
4-y degree or more	134 (34)	196 (47)		102 (35)	154 (46)	
Missing	4 (1)	3 (1)		2 (1)	2 (1)	
Marital Status			0.79			0.91
Married/living as married	320 (80)	341 (82)		236 (81)	272 (81)	
Other	75 (19)	70 (17)		53 (18)	60 (18)	
Missing	3 (1)	3 (1)		2 (1)	2 (1)	
Family history			0.95			0.85
Yes	45 (11)	51 (12)		34 (12)	40 (12)	
No/do not know	348 (87)	353 (85)		253 (87)	285 (85)	
Missing	5 (1)	10 (2)		4 (1)	9 (3)	
Previous PSA			0.60			0.69
Yes	177 (44)	196 (47)		127 (44)	161 (48)	
No	88 (22)	89 (21)		62 (21)	72 (22)	
Missing	133 (33)	129 (31)		102 (35)	101 (30)	
Screening preference			0.79			0.88
Want to be screened	315 (79)	325 (79)		225 (77)	264 (79)	
Does not want to be screened	33 (8)	40 (10)		29 (10)	32 (10)	
Undecided	46 (12)	46 (11)		34 (12)	36 (11)	
Missing	4 (1)	3 (1)		3 (1)	2 (1)	

* P value for test of whether persons who completed both surveys were the same as those who only participated in the baseline survey.

$^\dagger P$ value for test of whether intervention and comparison groups have similar compositions in follow-up cohort.

who had a prior PSA were more likely to be decided, to show decisional consistency, and in addition, had higher levels of decisional conflict.

Table 3 presents changes in primary and secondary outcomes between baseline and follow-up across intervention and comparison sites, in bivariate and adjusted

analyses. In these analyses, relative to the comparison group, there was a greater percentage of men in the intervention group who were undecided about screening at baseline and had made a screening decision at follow-up (21% versus 13%, respectively; $P = 0.01$). Although not an outcome of the study, 77% to 79% of men in both

Table 2. Bivariate regression results for association between sociodemographic variables and outcomes

Characteristic	Decisional status OR (95% CI)	CaP knowledge Regression coefficient (95% CI)	Decision self-efficacy Regression coefficient (95% CI)	Decisional consistency OR (95% CI)	Control preferences OR (95% CI)	Decisional conflict Regression coefficient (95% CI)
Age						
45-49	(Reference)	(Reference)	(Reference)	(Reference)	(Reference)	(Reference)
50-54	1.23 (0.80-1.89)	-0.98 (-3.59-1.64)	2.88 (-0.15-5.92)	1.63 (0.82-3.25)	0.63 (0.23-1.70)	3.32 (-0.58-7.21)
≥55	1.42 (0.84-2.14)	-1.52 (-4.20-1.16)	1.32 (-2.43-5.08)	2.37 (0.98-5.20)	0.65 (0.28-1.48)	9.04 (5.18-12.89)*
Race/ethnicity						
White, non-Hispanic	0.86 (0.52-1.42)	-0.19 (-4.07-3.70)	-7.35 (-13.44--1.26) [†]	0.63 (0.63-2.94)	1.38 (0.41-4.58)	4.80 (-0.92-10.57)
Other	(Reference)	(Reference)	(Reference)	(Reference)	(Reference)	(Reference)
Household income						
<\$50,000	(Reference)	(Reference)	(Reference)	(Reference)	(Reference)	(Reference)
\$50,000-74,999	1.18 (0.76-1.83)	0.49 (-2.90-3.88)	-2.33 (-7.81-3.15)	1.07 (0.44-2.63)	2.10 (0.80-5.55)	-3.13 (-9.61-3.35)
≥\$75,000	1.60 (0.96-2.69)	0.88 (-1.56-3.31)	-5.41 (-11.00-0.16)	0.99 (0.52-1.87)	1.72 (0.58-5.09)	-0.07 (-6.60-6.45)
Education						
High school or less	0.52 (0.36-0.76)	1.20 (-3.05-5.45)	1.89 (-2.62-6.40)	1.23 (0.94-1.61)	0.48 (0.21-1.10)	1.68 (-3.33-6.69)
Some college 4-y degree or more	0.68 (0.47-0.98) [†] (Reference)	1.41 (-0.73-3.56) (Reference)	-0.76 (-3.73-2.22) (Reference)	0.98 (0.58-1.66) (Reference)	0.56 (0.31-1.02) (Reference)	2.94 (-1.34-7.21) (Reference)
Marital status						
Married/living as married	0.86 (0.47-1.59)	0.83 (-2.96-4.62)	-0.01 (-4.16-4.14)	1.15 (0.72-1.83)	0.60 (0.33-1.09)	3.11 (-7.90-4.30)
Other	(Reference)	(Reference)	(Reference)	(Reference)	(Reference)	(Reference)
Family history						
Yes	1.55 (0.81-2.99)	2.19 (-1.84-6.22)	-0.68 (-4.71-3.34)	2.48 (0.87-6.80)	1.82 (0.88-3.76)	0.40 (-5.11-5.90)
No/do not know	(Reference)	(Reference)	(Reference)	(Reference)	(Reference)	(Reference)
Previous PSA						
Yes	2.57 (1.72-3.85)*	0.39 (-2.73-3.51)	-2.52 (-5.65-0.61)	2.04 (1.18-3.51) [†]	1.76 (0.64-4.89)	7.89 (4.25-11.54)*
No	(Reference)	(Reference)	(Reference)	(Reference)	(Reference)	(Reference)

Abbreviations: OR, odds ratio; 95% CI, 95% confidence interval.

* $P < 0.01$.[†] $P < 0.05$.

groups preferred CaP screening at both time points. Only 23% of men changed in their screening preferences between baseline and follow-up, and this did not vary by intervention arm. Approximately equal percentages of men in both groups changed from preferring to be screened, to preferring not to be screened (56 men changed from undecided/against screening to want screening, and 66 men changed the preferences in the reverse).

Men in the intervention group experienced greater improvements in knowledge scores than men in the comparison group; more than half (54%) of men in intervention sites had improved knowledge scores versus 39% of men in comparison sites. The average increase was 10 percentage points in the intervention group versus 4 percentage points in the comparison group ($P = 0.03$). There were no discernable changes in mean decision self-efficacy or decisional consistency scores in either group across time points.

Changes in secondary outcomes by treatment group

Overall, at both time points across intervention and comparison sites, the majority of men (91-95%) wanted an active or collaborative role in decision making. Preferred role in decision making did not change significantly between baseline and follow-up in either intervention or comparison sites. Mean scores in decisional conflict were low in intervention and comparison groups at baseline (mean scores, 25 versus 18, respectively). In multivariate analyses, men in the intervention group had marginally reduced decisional conflict scores ($P = 0.09$).

Subgroup analyses among DA users

Across the six intervention sites, 30% of the cohort reported using the DA (86 of 291). However, because we used an "intention to treat" model, use of the DA was *not* required for cohort membership. According to our process-tracking system, overall use of the tool among *all* age-eligible men in intervention work sites ranged from 23% to 59% (total $n = 335$ of a possible 1,118 eligible men across sites). DA users did not differ significantly by sociodemographic or health characteristics compared with nonusers (data not shown).

Men who used the DA were significantly more likely to have made a decision about screening at the follow-up survey, compared with nonusers (35% versus 14%, $P < 0.01$; Table 4). In addition, 79% of users improved their knowledge scores, whereas only 43% of nonusers had improved knowledge at the time of the follow-up survey (+12% points versus +1% points; $P < 0.01$). DA users also had a greater sense of self-efficacy in decision making; 41% of men in the intervention group versus 38% in the comparison group improved in their decision self-efficacy score ($P < 0.01$ in bivariate analyses). Nonetheless, decisional consistency was not different between the two groups. Although there were no significant differences between users and nonusers in preferences for control in decision making, DA users experienced a significant decrease in decisional conflict, compared with non-DA users ($P = 0.03$).

When asked about their satisfaction with the DA, 67% reported that the tool was "very helpful" in making screening decisions, and 70% of users were "very" or "mostly" satisfied with the DA. Importantly, 71% reported that the main message was that "men need to make an individual decision about CaP screening with their medical providers." However, 26% thought the main message was that "men should get an annual PSA test."

Discussion

Access to a computer-tailored DA over a 3-month period at the workplace resulted in a significant increase in the proportion of men who made a decision about CaP screening. In addition, the intervention produced a significant improvement in CaP knowledge, yet mean knowledge scores remained fairly low in both intervention and control groups. Neither decision self-efficacy nor decisional consistency was measurably improved. Yet, men in the intervention group reported marginally reduced decisional conflict. Unlike some prior studies, we used an "intention-to-treat" model in the primary analyses. In subgroup analyses among DA users, men were much more likely to have made a screening decision, have striking improvements in their levels of knowledge, and have increased decision self-efficacy.

Our findings align with some previous IDM trials. The previously cited review found that of 18 trials, 14 reported improvements in knowledge (11). The effect of DAs on decision self-efficacy has received far less attention; only two prior studies assessed this construct, each using one item for assessment. Gattellari et al. (31) found that patients who received DAs had greater confidence in their decision compared with patients who received a leaflet describing CaP screening risks and benefits. In contrast, Frosch et al. (32) reported reduced confidence in screening decisions among DA users compared with individuals in the control condition. Reductions in decisional conflict following DA use have been shown in several studies (31, 33-35). Six (13, 35-39) of nine CaP DA studies (31, 34, 37-43) have also produced a decreased preference for screening, although this was not the case among men in our study. We have been unable to locate any DA trials that evaluated consistency of decision making with one's values; our study found no measurable change in decisional consistency. Only four prior studies have evaluated computer-based DAs (13-16). Of these, two showed improved knowledge (14, 15) and one reported increased desire for decisional control (14). In terms of screening preferences, results have been mixed, with some finding a decreased desire to be screened and others finding an increased desire to be screened (13, 14, 16). Of note, only one of these interventions was tailored to individual user characteristics (although only on the basis of family history; ref. 15).

Before a discussion of implications, limitations of our study must be acknowledged. First, overall use of the DA among men in the intervention group was lower

Table 3. Changes in primary and secondary outcomes between baseline and follow-up surveys, by intervention arm, Take the Wheel Trial (*n* = 625)

Primary outcomes	Intervention (<i>n</i> = 291)		Comparison (<i>n</i> = 334)		Bivariate regression	Adjusted regression
	Baseline	Follow-up	Baseline	Follow-up	Coefficient or OR, (95% CI), <i>P</i>	Coefficient or OR, (95% CI), <i>P</i>
Decisional status (%)						
Decided	31%	43%	40%	43%		
Undecided	69	57	60	57		
Missing	0	0	0	0		
<i>Changes between baseline and follow-up*</i>						
Remained decided		22%		30%	1.26	1.53
Remained undecided		49		48	(0.99-1.61)	(1.15-2.05)
Became decided		21		13	0.07	<0.01
Became undecided		8		9		
CaP knowledge [†]						
Mean	56	66	56	60	3.25	4.24
SEM	2.68	2.08	1.23	1.6	(-0.52-7.02)	(0.40-8.07)
% of men with improved score		54%		39%	0.09	0.03
Decision self-efficacy [†]						
Mean	83	83	79	79	0.17	1.35
SEM	2.46	2.36	1.59	1.81	(-2.70-3.03)	(-1.24-3.93)
% of men with improved score		39%		40%	0.91	0.31
Decisional consistency (%)						
Consistent	71%	69%	73%	74%		
Inconsistent	28	31	26	24		
Missing	1	0.3	1	2		
<i>Changes between baseline and follow-up[§]</i>						
Remained consistent		53%		58%	0.73	
Remained inconsistent		13		10	(0.48, 1.12)	
Became consistent		15		16	0.15	
Became inconsistent		17		13		

(Continued on the following page)

Table 3. Changes in primary and secondary outcomes between baseline and follow-up surveys, by intervention arm, Take the Wheel Trial (*n* = 625) (Cont'd)

Secondary outcomes	Intervention (<i>n</i> = 291)		Comparison (<i>n</i> = 334)		Bivariate regression	Adjusted regression
	Baseline	Follow-up	Baseline	Follow-up	Coefficient or OR, (95% CI), <i>P</i>	Coefficient or OR, (95% CI), <i>P</i>
Control preferences (%)						
Active/collaborative	94%	95%	91%	92%		
Passive	5	4	8	7		
Missing	1	0.3	0.3	1		
<i>Changes between baseline and follow-up</i>						
Did not change		92%		87%	1.36	
Change to passive		3		5	(0.66, 2.82)	
Change to active/collaborative		3		7	0.41	
Decisional conflict ^{††}						
Mean	25	14	28	20	−3.33	−2.06
SEM	2.08	2.01	2.12	2.07	(−6.69, 0.02)	(−4.42, 0.29)
% of men with improved score		53%		49%	0.05	0.09

NOTE: Bivariate and adjusted analyses control for work site cluster.

**Change in decisional status*: Adjusted model is a logistic regression model with outcome being follow-up decisional status. Adjusted results indicate the odds of being decided at follow-up in the intervention cohort relative to the comparison cohort, controlling for baseline decisional status, age, education, and previous PSA.

†*Change in CaP knowledge*: Adjusted model is a linear regression model with outcome being the difference between follow-up and baseline knowledge. Adjusted results indicate the average change in knowledge between follow-up and baseline comparing intervention and comparison cohorts, controlling for age, race, income, education, marital status, family history of prostate cancer, and previous PSA.

‡*Change in decision self-efficacy*: Adjusted model is a linear regression model with outcome being the difference between follow-up and baseline decision self-efficacy. Adjusted results indicate the average change in decision self-efficacy between follow-up and baseline comparing intervention and comparison cohorts, controlling for age, race, income, education, marital status, family history of prostate cancer, and previous PSA.

§*Change in decisional consistency*: Adjusted analysis not estimable due to sample size constraints. Unadjusted results indicate the odds of being consistent at follow-up in the intervention cohort relative to the comparison cohort, controlling for baseline decisional consistency.

||*Change in control preferences*: Adjusted analysis not estimable due to sample size constraints. Unadjusted results indicate the odds of preferring an active or collaborative relationship at follow-up in the intervention cohort relative to the comparison cohort, controlling for baseline control preferences.

††*Change in decisional conflict*: Adjusted model is a linear regression model with outcome being the difference between follow-up and baseline decisional conflict. Adjusted results indicate the average change in decisional conflict between follow-up and baseline comparing intervention and comparison cohorts, controlling for age, race, income, education, marital status, family history of prostate cancer, and previous PSA.

Table 4. Changes in primary and secondary outcomes between baseline and follow-up surveys in intervention work sites, by DA use, Take the Wheel Trial (*n* = 289)

Primary outcomes	Intervention (<i>n</i> = 86)		Comparison (<i>n</i> = 203)		Bivariate regression	Adjusted regression
	Baseline	Follow-up	Baseline	Follow-up	Coefficient or OR, (95% CI), <i>P</i>	Coefficient or OR, (95% CI), <i>P</i>
Decisional status (%)						
Decided	28%	57%	32%	37%		
Undecided	72	43	68	63		
Missing	0	0	0	0		
<i>Changes between baseline and follow-up*</i>						
Remained decided		22		23	2.93	
Remained undecided		37		54	(1.49-5.79)	
Became decided		35		14	<0.01	
Became undecided		6		9		
CaP knowledge [†]						
Mean	59	73	62	64	13.34	10.98
SEM	3.04	2.44	3.60	3.43	(10.09-16.58)	(6.56-15.39)
% of men with improved score		79%		43%	<0.01	<0.01
Decision self-efficacy [†]						
Mean	80	77	79	76	3.30	-2.53
SEM	4.36	3.65	4.04	2.95	(0.76-5.84)	(-1.24-6.30)
% of men with improved score		41%		38%	0.01	0.19
Decisional consistency (%)						
Consistent	77%	70%	69%	68%		
Inconsistent	22	30	30	31		
Missing	1	0	1	0.5		
<i>Changes between baseline and follow-up[§]</i>						
Remained consistent		56%		53%	1.01	
Remained inconsistent		8		15	(0.50-2.04)	
Became consistent		14		15	0.98	
Became inconsistent		21		16		

(Continued on the following page)

Table 4. Changes in primary and secondary outcomes between baseline and follow-up surveys in intervention work sites, by DA use, Take the Wheel Trial (*n* = 289)

Secondary outcomes	Intervention (<i>n</i> = 86)		Comparison (<i>n</i> = 203)		Bivariate regression	Adjusted regression
	Baseline	Follow-up	Baseline	Follow-up	Coefficient or OR, (95% CI), <i>P</i>	Coefficient or OR, (95% CI), <i>P</i>
Control preferences (%)						
Active	58%	67%	70%	67%		
Collaborative	34	28	25	29		
Passive	6	5	4	4		
Missing	2	0	1	0.5		
Changes between baseline and follow-up						
Did not change		62%		70%	0.98	
Change to passive		16		16	(0.26-3.62)	
Change to active/collaborative		20		12	0.97	
Decisional conflict [¶]						
Mean	29	13	30	22	-14.77	-8.29
SEM	2.73	3.54	2.26	2.32	(-24.55 - -4.99)	(-15.54 - -1.04)
% of men with improved score		59%		45%	<0.01	0.03

NOTE: Bivariate and adjusted analyses control for work site cluster.

*Change in decisional status: Adjusted analysis not estimable due to sample size constraints. Unadjusted results indicate the odds of being decided at follow-up among DA users relative to the nonusers, controlling for baseline decisional status.

†Change in CaP knowledge: Adjusted model is a linear regression model with outcome being the difference between follow-up and baseline knowledge. Adjusted results indicate the average change in knowledge between follow-up and baseline comparing DA users and nonusers, controlling for age, race, income, education, marital status, family history of prostate cancer, and previous PSA.

‡Change in decision self-efficacy: Adjusted model is a linear regression model with outcome being the difference between follow-up and baseline decision self-efficacy. Adjusted results indicate the average change in decision self-efficacy between follow-up and baseline comparing DA users and nonusers, controlling for age, race, income, education, marital status, family history of prostate cancer, and previous PSA.

§Change in decisional consistency: Adjusted analysis not estimable due to sample size constraints. Unadjusted results indicate the odds of being consistent at follow-up among DA users relative to nonusers, controlling for baseline decisional consistency.

||Change in control preferences: Adjusted analysis not estimable due to sample size constraints. Unadjusted results indicate the odds of preferring an active or collaborative relationship at follow-up among DA users relative to nonusers, controlling for baseline control preferences.

¶Change in decisional conflict: Adjusted model is a linear regression model with outcome being the difference between follow-up and baseline decisional conflict. Adjusted results indicate the average change in decisional conflict between follow-up and baseline comparing DA users and nonusers, controlling for age, race, income, education, marital status, family history of prostate cancer, and previous PSA.

than we had hoped. Prior research has documented low rates of health program participation among males and “blue collar workers” (44). National data show that only 9% of eligible workers participate in available work site wellness programs (44). Thus, the participation rate among eligible men in this study should not have been entirely unexpected, particularly given that CaP is a highly personal topic. Moreover, given production demands in manufacturing work sites, it may have been difficult for men to take time away from their jobs, despite management agreement to let them do so. Still, improving rates of participation is an important goal for future efforts. Second, findings from this sample of predominantly white, employed men may not be generalizable to men of varied racial/ethnic backgrounds or men who are not employed. In particular, African-American men, who are at high risk for CaP, may have unique information needs and concerns (45) that were not addressed here. Given the disproportionate burden of CaP borne by African-American men, development and testing of CaP screening interventions specifically for this audience should be a priority. Third, we used a previously unvalidated measure of decisional consistency. Existing strategies to assess values are based on probability assessments and require a high level of numeracy (27). Although we implemented several strategies (e.g., cognitive testing) to enhance the face and content validity of the measure we developed, additional work to fully evaluate its psychometric properties is needed. In addition, given the high scores on this scale across the sample, a more sensitive measure of values may be needed. Fourth, we used self-reported PSA, which can underreport rates of screening (46). We did find a substantial amount of missing data on PSA history. However, our primary aims were to examine IDM outcomes, not screening participation. Finally, we examined only short-term changes in IDM after a 3-month period. Because PSA screening may be offered annually, it is important to examine how various aspects of decision making change over time.

Nevertheless, this study makes important contributions to the existing literature on interventions to promote IDM for CaP. First, we examined multiple aspects of IDM, as defined by the U.S. Preventive Services Task Force. Prior studies have tended to focus on knowledge, and there has been insufficient study of other features of IDM (47). It is important that interventions assess all aspects of IDM because knowledge is not the sole criterion for high-quality decision making. Second, we report results from a computer-tailored DA. To date, the vast majority of DAs have been nontailored and have taken the form of videos or written materials. As noted, there are numerous advantages to using computer technology to deliver tailored IDM messages. Finally, we present data from an intervention offered in the work setting. To our knowledge, this is the first trial of a workplace intervention directed at IDM for CaP. Most interventions have been conducted in clinical settings despite the call for providing interventions in community settings (10).

The results of this study point to several important issues for consideration in future interventions. First, use of the DA resulted in an increase in decision making, improved knowledge, and enhanced self-efficacy. However, decisional consistency was consistently high and unaffected. This may have been due to “ceiling effects” in the measurement of this variable. Alternatively, it may be that many men overestimate the efficacy of screening and their providers' endorsement of PSA, leading them to disregard the potential disadvantages and harms (33, 48). We found that men are confident in their decision-making capabilities and are making decisions consistent with their values, even in the absence of what most would consider adequate knowledge. Are these truly informed decisions? Future research should consider whether it would be reasonable to identify a required level of knowledge about the benefits, risks, and limitations of CaP screening as a prerequisite for IDM, such as the case for informed consent (18).

Second, more research is needed on how to create “balanced” messages in the context of IDM interventions. Communicating the uncertain balance between potential benefits (e.g., early detection) and harms (e.g., risk of overtreatment) of CaP screening presents a major challenge. In our study, nearly a quarter of DA users thought the main message was that “men should be screened annually.” Other IDM studies have similarly reported that men interpreted educational messages as promoting screening (49). There is ample evidence that message framing and varied formats for presenting risk information can affect decisions (50). Experimental manipulation of these factors may be informative for development of future DAs.

Based on these findings, we conclude that a computer-tailored DA offered at the workplace can be effective in promoting IDM, but participation among male, blue collar workers in this study was suboptimal. A recent review of health promotion programs offered in the work site found that less than 10% of employees accessed program components required for successful interventions (44). It is clear that better strategies for engaging men, particularly those of diverse racial/ethnic backgrounds, are needed. Work sites are a logical venue for reaching men, and many already offer health promotion programs (44). Integrating health topics, such as IDM for CaP screening, into existing programs that are more highly used (e.g., blood pressure screening) could potentially enhance programmatic dose and reach. Framing interventions in terms of “men's health” rather than “cancer screening” may also draw a larger audience. Efforts might include engaging men in more socially relevant or trusted environments (e.g., barbershops, sporting events) where they may be more likely to discuss health issues with one another. Given the effect of the DA among those who used it, we are optimistic that using interactive state-of-the-art technology will help to better engage men and will facilitate the ultimate goal of preparing them to participate in complex decisions about their medical care.

Appendix 1

Values questions	Would choose to be screened* (n = 489)	Would choose NOT to be screened* (n = 131)
It is important to me to have a PSA test, even if doctors are not sure that screening can save lives.	92%	61%
It is important to me to have a PSA test, even if there is a chance the results could be wrong.	91%	54%
Finding prostate cancer early and getting treatment is worth any possible side effect, including difficult having sex or leaking urine.	91%	76%
If I had prostate cancer, I would want to know—even if it wasn't going to kill me.	97%	93%
I would not want to have a PSA test unless doctors are reasonably sure that it can save lives.	18%	53%
I prefer not to be screened for prostate cancer if there is a chance the results could be wrong.	11%	39%
If getting treated for prostate cancer means that I wouldn't be able to have sex or that I might not be able to control my urine, I might choose not to get screened.	24%	93%
If I had prostate cancer, I would rather not know—especially if it wasn't going to kill me.	10%	19%

NOTE: Negative items were reverse coded, such that high scores indicate stronger, more positive values about screening.

*Percent strongly agree or agree.

Disclosure of Potential Conflicts of Interest

No potential conflicts of interest were disclosed.

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